

DEVANSHU SINGH

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Education

Indian Institute of Information Technology Bhagalpur

Aug 2023 – July 2027

Bachelor of Technology in Mechatronics and Automation Engineering

Bhagalpur, Bihar

Grade: 9.15 (latest), 8.16 (overall)

Skills

Languages: C, C++, Python, SQL

Frameworks & Tools: FastAPI, LangChain, LangGraph, PyTorch, TensorFlow, Keras, Git, GitHub, Jupyter

Databases: MySQL, PostgreSQL, MongoDB, ChromaDB, FAISS

CS Fundamentals: Data Structures, Algorithms, OOPs, DBMS

AI/ML: Machine Learning, Deep Learning, NLP, LLMs, RAG, AI Agents, Prompt Engineering

Experience

Indian Institute of Technology Patna 🌐

Jun 2025 – Nov 2025

ML Research Intern

Patna, Bihar

- Applied machine learning techniques to analyze quantum mechanical data and predict magnon blockade phenomena using supervised learning models.
- Engineered and optimized feature extraction pipelines from quantum simulation datasets to enhance model interpretability and accuracy.
- Performed model development, experimentation and result analysis using Python, PyTorch, and deep learning frameworks while strengthening expertise in quantum theory fundamentals.

Projects

Invoicely: AI-Powered Invoice Agent | *LangGraph, FastAPI, ChromaDB, SQLite, LlamaParse* | 🌐

- Developed a dual-engine Agentic RAG system using LangGraph that routes semantic queries to ChromaDB and deterministic financial calculations to a read-only SQLite database for reliable numerical reasoning.
- Built an asynchronous document ingestion pipeline using FastAPI, LlamaParse, and Gemini to extract structured invoice data, with Pydantic-based validation and automatic retry on schema validation failures before database insertion.
- Improved agent reliability by implementing retry limits for Text-to-SQL generation, domain-specific query boundaries, and graceful fallback responses for failed SQL execution while tracing workflows with LangSmith.

Intelligent Crop Recommendation API | *LangChain, LLM, FastAPI, FAISS, Pydantic* | 🌐 🌐

- Engineered a RESTful Retrieval-Augmented Generation (RAG) API using FastAPI and LangChain Expression Language (LCEL) to generate location-aware crop recommendations from heterogeneous agricultural data sources.
- Orchestrated a hybrid retrieval pipeline combining real-time weather and soil information with static environmental, market, and domain knowledge indexed in a FAISS vector store to deliver contextually relevant recommendations.
- Integrated Google Gemini with structured Pydantic output parsing to enforce schema-compliant JSON responses, improving API reliability and downstream consumption.

Achievements

- Finalist at multiple national hackathons, including **HackTron** (IIIT Gwalior, **Top 0.1%**), **CodeSlayer** (NIT Delhi, **Top 0.02%**), and **Bit By Bit** (IIIT Bhagalpur, **Top 5**).
- Completed the **Machine Learning Specialization** by Stanford University and DeepLearning.AI. 🌐
- Completed IBM's **Exploratory Data Analysis for Machine Learning** certification with **Honors**. 🌐

Extracurricular

- **TEDx IIIT Bhagalpur:** Served as **Production Lead** in the organizing team, coordinating production workflows and ensuring seamless execution of the flagship TEDx event.
- **Unnati Welfare Society:** Served as Treasurer of an NGO providing education to **1000+** underprivileged children across Bhagalpur.
- Mentored and taught **200+** students across multiple government schools, promoting digital literacy and holistic learning.